TRENESIS

CARDIO TRAINING GUIDE

Evidence-based fundamentals for building real, lasting results.

NO FLUFF · NO PSEUDOSCIENCE · JUST RESULTS

BEFORE YOU START

Disclaimer & How to Use This Guide

This guide is for informational and educational purposes only. It does not constitute medical advice. Consult a qualified healthcare professional before starting any new exercise program, especially if you have pre-existing injuries or health conditions.

The information presented is based on current evidence-based exercise science. Individual results vary based on consistency, nutrition, sleep, and genetics.

How to Use This Guide

- Read Chapters 1–3 before picking a program. The principles make everything else click.
- Choose the program in Chapter 4 that fits your schedule and current level.
- Log every session – duration, distance, and heart rate (if you have a tracker). Progress requires data.
- Come back to Chapter 5 when you plateau. Most stalls are lifestyle issues.

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CHAPTER 01

Why Cardio?

The science-backed case for training your engine.

The Case for Cardiovascular Training

Cardiovascular training is one of the most powerful interventions for long-term health. The evidence is unambiguous: consistent cardio reduces all-cause mortality, lowers resting heart rate, improves insulin sensitivity, enhances mental health, and increases VO2 max – one of the strongest predictors of longevity ever measured.

Yet most people either skip cardio entirely or do it wrong – endless slow treadmill sessions that produce minimal adaptation. This guide fixes that. You will learn the fundamentals that produce real cardiovascular fitness across all goals.

Key Benefits

Heart Health	Lowers resting heart rate and blood pressure over time.
VO2 Max	More oxygen delivered to muscles – you get fitter at everything.
Fat Metabolism	Trained aerobic systems burn more fat at rest and during exercise.
Mental Health	Consistent cardio significantly reduces depression and anxiety
Metabolic Health	Improves insulin sensitivity and blood glucose regulation.
Longevity	VO2 max is among the strongest predictors of healthy ageing.

KEY INSIGHT

A VO2 max increase of just 3.5 ml/kg/min – roughly one MET – is associated with a 10–15% reduction in all-cause mortality risk. That is achievable in 8–12 weeks of consistent training.

NOTE

Many lifters associate cardio mainly with weight loss, but its benefits go far beyond burning calories. In the right amount, cardiovascular training can improve stamina, recovery, and overall training performance, making strength workouts more effective rather than interfering with them.

CHAPTER 02

Types of Cardio Workouts

Choose the modality that matches your goals and your lifestyle.

Steady-State Cardio (Zone 2)

Steady-state cardio means maintaining a consistent, moderate pace for a sustained period – typically 20–60 minutes. At Zone 2 intensity you should be able to hold a full conversation. This is the foundation of all cardio programming and is the most researched form of aerobic exercise. Activities include running, cycling, swimming, walking, rowing, and the elliptical.

High-Intensity Interval Training (HIIT)

HIIT alternates short bursts of near-maximal effort with periods of rest or easy movement. A common structure is 6×2 minutes hard with 2 minutes easy recovery. HIIT is time-efficient and produces strong cardiovascular adaptations, but it is demanding – limit HIIT to one or two sessions per week to allow full recovery.

Tempo / Threshold Training

Tempo runs or threshold intervals are performed at Zone 4 – comfortably hard, breathing heavy, but sustainable for 20–40 minutes. This zone develops your lactate threshold, allowing you to sustain higher speeds for longer. A classic structure is 4×4 minutes hard with 3 minutes easy recovery.

Long Slow Distance (LSD)

Long slow distance sessions are extended Zone 2 efforts – typically 45–90 minutes. They build aerobic base, fat-burning capacity, and mental resilience. One LSD session per week is a cornerstone of most endurance programs.

Active Recovery

Zone 1 movement – easy walking, light cycling, gentle swimming – is not a workout but a recovery tool. It increases blood flow, clears metabolic waste, and accelerates adaptation without adding training stress. Aim for 20–40 minutes on rest days.

> **KEY INSIGHT**
>
> Modality matters less than consistency: running, cycling, rowing, and swimming all build similar cardiovascular fitness. Pick what you can stick with long-term. Only prioritize specificity if performance in a particular sport matters (e.g., run to get better at running).

CHAPTER 03

Heart Rate and Training Zones

Understanding intensity is the key to training smarter.

Calculating Your Maximum Heart Rate

Heart rate zones define intensity. Your maximum heart rate (MHR) can be estimated as 220 minus your age. Zones are percentages of that number. A heart rate monitor removes the guesswork entirely and is one of the most valuable investments a cardio athlete can make.

ZONE	% OF MHR	FEEL	PURPOSE
Zone 1	50–60%	Very easy	Active recovery
Zone 2	60–70%	Conversational	Aerobic base, fat metabolism
Zone 3	70–80%	Comfortably hard	Aerobic threshold
Zone 4	80–90%	Hard, breathing heavy	Lactate threshold
Zone 5	90–100%	Maximum effort	VO2 max, anaerobic capacity

The 80/20 Rule

Decades of research on elite endurance athletes consistently shows the same pattern: roughly 80% of training volume at low intensity (Zone 1–2) and 20% at high intensity (Zone 4–5). Spending too much time in Zone 3 – the “grey zone” – is one of the most common and costly mistakes. It is too hard to allow full recovery, but not hard enough to maximise adaptation.

> **KEY INSIGHT**
>
> Zone 2 training builds the aerobic base that makes all other zones possible. If you cannot hold a conversation during an easy run, you are going too hard. Slow down – the adaptations still happen.

CHAPTER 04

Cardio Training Plans

Three complete programs. Pick one and commit.

Each program is evidence-based, time-efficient, and built around the principles in earlier chapters. Pick based on your schedule and current level.

program A

3-Day Aerobic Base Builder

Three sessions per week of Zone 2 cardio. Ideal for beginners or anyone returning after a break – and it also works very well alongside strength training programs. The priority is building a broad aerobic foundation. Choose any low-impact modality you enjoy: walking, cycling, swimming, or easy running.

Sample Week:

- Monday: 30 min Zone 2
- Wednesday: 30 min Zone 2
- Friday: 35 min Zone 2

program B

4-Day Endurance Build

Four sessions per week – three aerobic base sessions and one tempo run or threshold interval session. This structure accelerates aerobic development while introducing controlled stress at higher intensities.

Sample Week:

- Monday: 35 min Zone 2
- Wednesday: 25 min Zone 4 intervals (4×4 min hard, 3 min easy)
- Friday: 40 min Zone 2
- Sunday: 45 min Zone 2 long session

program C

5-Day Performance Program

Five sessions per week for those with a solid base. Uses the 80/20 structure explicitly: four easy days and one hard interval session. Well suited for improving a 5K or 10K time, or for athletes wanting strong cardiovascular conditioning alongside strength training.

Sample Week:

- Monday: 40 min Zone 2
- Tuesday: 30 min Zone 2
- Wednesday: HIIT (6×2 min Zone 5, 2 min walk)
- Friday: 45 min Zone 2
- Saturday: 50 min Zone 2 long run

CHAPTER 05

Common Mistakes

The errors that stall 90% of lifters.

Training in the Grey Zone

Doing most sessions at moderate intensity – not easy enough to allow full recovery, not hard enough to maximise adaptation. Follow the 80/20 rule. Most sessions should feel genuinely easy.

Not Tracking Progress

If you do not log duration, distance, and heart rate, you cannot know if you are improving. Weekly data reveals trends that feel invisible day-to-day.

Skipping Warm-Ups

Starting hard without a proper warm-up increases injury risk and reduces performance. Spend 5–10 minutes in Zone 1 and use dynamic stretching before any moderate or high-intensity session.

Ignoring Nutrition and Hydration

Cardio depletes glycogen and fluids. Sessions over 60 minutes require fuelling. Chronic under-fuelling leads to fatigue, poor adaptation, and muscle loss.

Only Doing Cardio

Cardio without any resistance training leads to muscle loss over time, especially in a calorie deficit. A minimum of two strength sessions per week protects muscle mass and improves running economy. For more information, check our **Strength Training Guide**.

Program Hopping

Switching programs every few weeks prevents adaptation. Cardiovascular changes take weeks to accumulate. Commit to a program for at least 8–12 weeks before evaluating results.

CHAPTER 06

Frequently Asked Questions

How long until I see results?

Cardiovascular improvements begin within 2–4 weeks (largely from increased stroke volume and blood volume). Meaningful VO2 max gains are typically visible after 6–8 weeks. Body composition changes from cardio alone take 8–16 weeks. Patience is not optional

Should I do cardio before or after lifting?

For most people, lift first. Fatigue from cardio impairs strength performance more than the reverse. If your primary goal is endurance, flip the order – or better, separate them by at least 6 hours or on different days entirely..

Will cardio kill my muscle gains?

Not if programmed correctly. Excessive high-intensity cardio on top of a full strength program can interfere with recovery. But 3–4 sessions of Zone 2 cardio per week will not meaningfully impact muscle growth – and it will improve your work capacity in the gym.

Running, cycling, rowing – does it matter?

For general cardiovascular fitness, the modality matters less than the intensity and consistency. Choose something you enjoy and can sustain. If sport-specific performance matters, specificity applies – run to improve running.

How do I know if I'm training hard enough?

For Zone 2, use the talk test: you should be able to hold a full conversation. For intervals, you should be unable to say more than a few words.

Is soreness a sign of a good cardio session?

Not really. Mild leg soreness after a new running stimulus is normal. But soreness is not a reliable measure of cardiovascular adaptation. What matters is whether your heart rate at the same pace is trending downward over weeks – that is adaptation.

Ready to put it into practice?

More free guides, training content, and resources at:

trenesisgym.github.io/trenesis/

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